The other points such as H, K, R and S lie on higher iso cost lines indicating that a larger outlay is required, which exceeds the financial resources of the firm.

3.11 Cobb-Douglas Production Function



W.Cobb and Paul H.Douglas

Cobb-Douglas Production Function is a specific standard equation applied to describe how much output can be made with capital and labour inputs. It is used in empirical studies of manufacturing industries and in inter-industry comparisons. The relative shares of labour and capital in total output can also be determined. It is still used in the analysis of economies of modern, developed and stable nations in the world.

The Cobb-Douglas Production Function was developed by Charles W. Cobb and Paul H. Douglas, based on their empirical study of American manufacturing industry. It is a linear homogeneous production function which implies that the factors of production can be substituted for one another up to a certain extent only.

The Cobb-Douglas production function can be expressed as follows.

$$Q = AL^{\alpha} K^{ß}$$

Where, Q = output; A = positive constant; K = capital; L = Labor α and β are positive fractions showing, the elasticity coefficients of outputs for the inputs labor and capital, respectively.

ß = (1- α) since α + ß = 1. denoting constant returns to scale.

Factor intensity can be measured by the ratio ß / α.

The sum of α + ß shows the returns to scale.

i) (α + ß) =1, constant returns to scale.

ii) (α + ß) <1, diminishing returns to scale.

iii) (α + ß) >1, increasing returns to scale.

- The production function explains that with the proportionate increase in the factors, the output also increases in the same proportion.
- Cobb-Douglas production function implies constant returns to scale.
- Cobb-Douglas Production Function is a specific standard equation applied to describe how much output can be made with capital and labour inputs. It is used in empirical studies of manufacturing industries and in inter-industry comparisons. The relative shares of labour and capital in total output can also be determined. It is still used in the analysis of economies of modern, developed and stable nations in the world.
- Cobb-Douglas production function considered only two factors like labour